Clock regularity and hybrid reconstruction for reflected-clock iterated Brownian motion

Project Stardust

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Abstract

Let $Z_t = X_{|Y_t|}$ for independent Brownian motions X, Y. We reconstruct the closed-operator argument that combines static recovery in the scenery variable with a predictable integral in the hidden Brownian clock. For absolutely summable multitime Fourier observations, the two channels recover the entire centered observable through the original joint intrinsic divergence. A stronger frequency summability condition puts the conditional source mean in the original clock derivative domain and identifies the clock integrand by predictable projection. Fold and collision cusps are treated by strong graph approximation, including their zero levels. Complete proofs separate ambient adjoints, observable projection, hidden-clock predictability and observed-process causality. Original-domain multipliers and an exact conditional-L1 criterion for the native adjoint are proved, retaining the full observation projection and all bounded-test hypotheses. A sine observation has zero source mean but positive variance, so clock-mean representation alone cannot reconstruct IBM. This is an exposition of verified results and classical Gaussian mechanisms, with no claim of completed physical-time IBM calculus or external novelty.

1 Scope, spaces and original operators

The frozen mathematical source is verified commit

`bab688bb6140cd6316c80680095a914ea0d346db`.

The accompanying theorem map records exact declaration headers and source hashes. No mathematical source or original research manuscript was changed. The proofs below reorganize the source into ordinary Gaussian Hilbert-space notation. Such reorganizations are distinguished in the map from literal formal declarations. In particular, the density argument for Brownian representation uses stochastic exponentials here, whereas the formal source uses its completed Brownian polynomial range.

Work on the product of two Gaussian probability spaces. The scenery X and clock Y are independent standard Brownian motions on $[0, \infty)$, with continuous versions. Write $\mathcal{K} = L^2(\Omega_X \times \Omega_Y)$ and

$$Z_t = X_{|Y_t|}, \qquad \mathcal{H} = L^2(\sigma(Z_t : t \geq 0)) \subseteq \mathcal{K}.$$

All equalities of random variables or fields are equalities of actual L^2 classes unless a representative is specified. Let $P : \mathcal{K} \to \mathcal{H}$ be orthogonal projection. Thus P is conditional expectation onto the full observed IBM sigma algebra, not onto an observed past.

Let D_X, D_Y be the original closed Gaussian derivatives on the product space, valued in the corresponding ambient Gaussian direction fields. Their Hilbert adjoints are δ_X, δ_Y. Directions for the two factors can be identified with separate copies of $L^2(\mathbb{R}_+)$. Define

$$\operatorname{Dom} D_{\mathcal{H}} = \mathcal{H} \cap \operatorname{Dom} D_X \cap \operatorname{Dom} D_Y, \qquad D_{\mathcal{H}} f = (D_X f, D_Y f).$$

The codomain is the direct sum of *ambient* field spaces. Fields are not assumed to be measurable functions of Z or predictable for its filtration. The original source proves that $D_{\mathcal{H}}$ is densely defined and closed. Density is a foundational input here: it follows there by proving that the annihilator of its native intrinsic core is zero. Closedness also follows directly by taking the intersection of the two closed partial graphs. Put $\delta_{\mathcal{H}} = D_{\mathcal{H}}^*$ with its actual adjoint domain.

Lemma 1.1 (Projected ambient adjoints)**.** *If $u \in \operatorname{Dom}\delta_X$ and $v \in \operatorname{Dom}\delta_Y$, then $(u, v) \in \operatorname{Dom}\delta_{\mathcal{H}}$ and*

$$\delta_{\mathcal{H}}(u, v) = P(\delta_X u + \delta_Y v).$$

This is an inclusion of domains; no equality with the full intrinsic adjoint domain is asserted.

Proof. For every $f \in \operatorname{Dom} D_{\mathcal{H}}$, ambient duality and $Pf = f$ give

$$\langle (u, v), D_{\mathcal{H}} f\rangle = \langle \delta_X u + \delta_Y v, f\rangle = \langle P(\delta_X u + \delta_Y v), f\rangle.$$

The last expression is bounded in $\|f\|_2$. This is exactly adjoint-domain membership, and density makes its representing vector unique. □

Source correspondence: `IBMJointIntrinsicDerivative jointDerivative_dense; ambient_pair_divergence; ambient_pair_duality`

One cannot omit P merely because f is intrinsic. In particular, $\mathbf{E}_X F$ and $F - \mathbf{E}_X F$ need not separately be intrinsic observations. Their sum can be intrinsic even when neither summand is.

2 Multipliers and exact conditional divergence domains

The projected adjoint also admits precise original-domain product rules. Abbreviate $D = D_{\mathcal{H}}$, $\delta = \delta_{\mathcal{H}}$ and $\mathcal{O} = \sigma(Z_t : t \geq 0)$. Through the original product/Fubini isometries, write its field space as $\mathcal{V} = L^2(\Omega; H_X \oplus H_Y)$, with pointwise real pairing $\langle \cdot, \cdot \rangle_{H_X \oplus H_Y}$. For an integrable scalar q, let

$$Cq = \mathbf{E}[q \mid \mathcal{O}] \in L^1.$$

This is an L^1 conditional expectation; the earlier Hilbert projection Pq is used only when its input belongs to L^2.

Bounded graph approximation and bounded separating tests

We first record the exact approximation facts needed for domain claims. The closed Gaussian chain rule gives

$$D\phi(v) = \phi'(v)Dv \qquad (v \in \operatorname{Dom} D)$$

when ϕ is bounded C^1 with continuous bounded derivative. It applies in each ambient Gaussian factor. The composed scalar remains $\mathcal{O}$-measurable, so the two graph identities belong to the original restricted joint domain. Here is the strong-graph mechanism: approximate in the original Gaussian graph by smooth cylindrical functions, apply the finite chain rule, and select an almost-everywhere convergent subsequence of the values. Bounded derivatives control the product with the converging gradients; dominated convergence controls $(\phi'(v_n) - \phi'(v))Dv$. Boundedness of ϕ' also gives convergence of the values. Closedness identifies the limit in each graph. Smooth approximations of ϕ on finite-dimensional cylinders give the same argument for C^1 functions. This is the usual Gaussian Sobolev chain mechanism, not an IBM physical-time chain rule.

Take $\chi_n(x) = n\tanh(x/n)$, $n \geq 1$. These are bounded smooth functions with $|\chi_n(x)| \leq |x|$, $|\chi_n'(x)| \leq 1$, $\chi_n(x) \to x$ and $\chi_n'(x) \to 1$. Thus, for every $v \in \operatorname{Dom} D$,

$$v_n = \chi_n(v) \longrightarrow v \text{ in } L^2, \qquad Dv_n = \chi_n'(v)Dv \longrightarrow Dv \text{ in } \mathcal{V}. \tag{1}$$

Both convergences follow by dominated convergence, using $|v|^2$ and $\|Dv\|^2$ respectively. In particular bounded domain elements form a strong graph core. The displayed hyperbolic truncations are a human choice of approximation; the source proves the same graph-core property with its own bounded smooth approximants.

A different fact is needed to separate L^1 densities. If $F \in \mathcal{H}$ and $|F| \leq 1$, density of $\operatorname{Dom} D$ gives $g_n \to F$ in L^2. Let $\chi(x) = \max(-1, \min(1, x))$. Convolve it with a nonnegative smooth unit-mass mollifier supported in $[-\varepsilon, \varepsilon]$. The result χ_ε is smooth, takes values in $[-1, 1]$, has derivative bounded by one and satisfies $\sup_x |\chi_\varepsilon(x) - \chi(x)| \leq \varepsilon$. The chain rule puts $u_n = \chi_{1/n}(g_n)$ in $\operatorname{Dom} D$, and

$$|u_n| \leq 1, \qquad \|u_n - F\|_2 \leq n^{-1} + \|g_n - F\|_2 \longrightarrow 0.$$

We used that clipping is nonexpansive and fixes F. The formal source uses the exact clipping identity $\chi(x) = (|x+1| - |x-1|)/2$ and its proved absolute-value chain rule; the smooth construction here supplies the same value approximation. No convergence of the derivatives of u_n is claimed in this statement.

Consequently an $\mathcal{O}$-measurable $a \in L^1$ satisfying $\mathbf{E}[va] = 0$ for every $v \in \operatorname{Dom} D$ with $|v| \leq 1$ must vanish. Apply the approximation to $F = \operatorname{sgn}(a)$, with sign zero at zero. An L^2-convergent sequence has an almost-everywhere convergent subsequence. Along it, $|u_n a| \leq |a|$, so dominated convergence gives $\mathbf{E}|a| = \lim_n \mathbf{E}[u_n a] = 0$. This proves $a = 0$ almost surely. The distinction between graph approximation and uniformly bounded value approximation is essential to the following proof.

Source correspondence: `IBMNativeBoundedGraphCore; IBMUniformBoundedNativeTests bounded_graph; uniformly_bounded_approximation`

Source correspondence: `IBMNativeL1TestSeparation integral_zero_on_bounded_observable; annihilator_zero`

An exact L^1-to-adjoint criterion

Theorem 2.1. *Suppose $U \in \mathcal{V}$ and $q \in L^1$ satisfy*

$$\langle U, Dv\rangle_{\mathcal{V}} = \mathbf{E}[vq] \quad \text{for every bounded } v \in \operatorname{Dom} D. \tag{2}$$

Then

$$U \in \operatorname{Dom}\delta \quad \Longleftrightarrow \quad Cq \in L^2.$$

When these conditions hold, $\delta U = Cq$ as an actual intrinsic L^2 class. For unbounded domain tests the valid full identity is $\langle U, Dv\rangle = \langle Cq, v\rangle_{L^2}$; integrability of vq is not asserted for arbitrary $v \in \operatorname{Dom} D$.

Proof. For every bounded observable v, conditional expectation gives $\mathbf{E}[vq] = \mathbf{E}[vCq]$, with both products integrable. If $U \in \operatorname{Dom}\delta$ and $y = \delta U$, adjoint duality and (2) imply $\mathbf{E}[v(Cq - y)] = 0$ for all bounded domain tests. The difference $Cq - y$ is observable and integrable, since the probability space has mass one and $y \in L^2$. The separating-test argument just proved gives $Cq = y$ almost surely, and therefore $Cq \in L^2$.

Conversely suppose $Cq \in L^2$. The bounded-test identity becomes $\langle U, Dv\rangle = \langle Cq, v\rangle$. For an arbitrary domain element use (1); convergence on both sides extends the identity to it. Thus $|\langle U, Dv\rangle| \leq \|Cq\|_2\|v\|_2$ on the whole domain, which is exactly membership in the Hilbert adjoint graph. Density of the original domain makes its output uniquely Cq. □

Source correspondence: `IBMConditionalL1Realization`
`conditional_pairing; full_pairing; realization_graph`

Source correspondence: `IBMConditionalAdjointDomain`
`domain_iff_conditional_L2; conditional_eq_adjoint`

Bounded products and the native divergence rule

If $f, v \in \operatorname{Dom} D$ are bounded, their product belongs to $\operatorname{Dom} D$ and

$$D(fv) = fDv + vDf.$$

To see this on the original graphs, choose a bounded smooth function agreeing with x^2 on a neighborhood of the common bounded ranges of f, v and $f + v$. The closed chain rule and $2fv = (f + v)^2 - f^2 - v^2$ prove the formula in each component. All terms are square integrable because both scalar factors are bounded.

Now let $f \in \operatorname{Dom} D$ be bounded and $U \in \operatorname{Dom} \delta$, without yet assuming an essentially bounded derivative of f. The actual density

$$q = f\delta U - \langle Df, U\rangle_{H_X \oplus H_Y} \tag{3}$$

belongs to L^1: the first product is in L^2, and Cauchy–Schwarz makes the field contraction integrable. Also $fU \in \mathcal{V}$. For a bounded domain test v, apply adjoint duality to fv:

$$\langle fU, Dv\rangle = \langle U, D(fv)\rangle - \mathbf{E}[v\langle Df, U\rangle] = \mathbf{E}[vq].$$

Theorem 2.1 therefore gives the exact rule

$$fU \in \operatorname{Dom} \delta \quad \Longleftrightarrow \quad C(f\delta U - \langle Df, U\rangle) \in L^2. \tag{4}$$

When it holds, the displayed conditional density is the adjoint value. Boundedness of f alone does not establish this remaining L^2 condition. One need not demand that the unprojected L^1 density itself be in L^2; the criterion concerns its actual conditional expectation.

Source correspondence: `IBMNativeProductOperators; IBMNativeWeakL1Divergence`
`bounded_leibniz; weak_pairing; weakDensity_integrable`

Source correspondence: `IBMConditionalAdjointDomain`
`weighted_domain_iff`

There is a useful sufficient multiplier class. Assume $|f| \le A$ almost surely and the combined field bound $\|Df(\omega)\| \le B$ almost surely, for finite $A, B \ge 0$. Then every $v \in \operatorname{Dom} D$, even unbounded, satisfies

$$\begin{aligned} fv \in \operatorname{Dom} D, \qquad & D(fv) = fDv + vDf, \\ \|D(fv)\|_2 \le A\|Dv\|_2 + B\|v\|_2. & \end{aligned} \tag{5}$$

Indeed use the bounded graph sequence v_n above. The bounded product rule applies to each fv_n; boundedness of f and Df gives convergence of its values and gradients to the displayed pair. Closedness of D proves membership and the triangle inequality proves the bound. The same calculation shows continuity under every original graph limit.

For every $U \in \operatorname{Dom}\delta$, (3) is now in L^2, so the weighted field automatically belongs to the adjoint domain and

$$\begin{aligned} \delta(fU) &= f\delta U - C\langle Df, U\rangle, \\ \|\delta(fU)\|_2 &\le A\|\delta U\|_2 + B\|U\|_2. \end{aligned} \tag{6}$$

We pulled out $f\delta U$ because it is already observable; conditional expectation is a contraction in L^2. In particular the contraction term cannot simply be assumed observable, since the derivative fields are ambient. The source states boundedness of the two derivative coordinates separately; the finite direct-sum identification gives the combined bound B used here. The printed constants are elementary human operator-norm bounds, not new named formal estimates.

Source correspondence: `IBMIntrinsicDomainMultiplier product_mem_domain; productElement_gradient_ae; productElement_graph_limit`

Source correspondence: `IBMNativeWeightedDivergence weighted_mem; weighted_value; full_pairing`

Nualart's primary notes [3], Proposition 5.2, equation (30), give the classical Gaussian divergence product rule on smooth cylindrical data; Proposition 5.4 states the closed Sobolev chain rule. The mechanism is classical. The present proof aligns it with the original restricted IBM graph and the precise conditional L^1 criterion, without claiming a new general product principle or priority for the restriction formulation. Here C always conditions on the *full* IBM observation sigma-algebra. These static multiplier and adjoint identities do not imply observed-past predictability, a physical-time stochastic integral or differentiability of the positive-time point observables excluded by the companion domain audit.

3 Static Gaussian recovery before predictability

We recall a Gaussian fact in a form that fixes all normalizations. Let D be the closed Gaussian derivative on one factor, $\delta = D^*$, and $L^2 = \bigoplus_{n\ge 0}\mathcal{C}_n$ its orthogonal Hermite-chaos expansion. For $f_n \in \mathcal{C}_n$, the original derivative has

$$\|Df_n\|_2^2 = n\|f_n\|_2^2, \qquad \delta Df_n = nf_n.$$

The identities follow first for finite products of normalized Hermite polynomials, by one-variable Gaussian integration by parts and orthogonality, and then by closure on each chaos. Derivative images from different chaos orders are orthogonal as direction fields.

Proposition 3.1 (Bounded original recovery)**.** *For $f = \sum_{n\ge 0} f_n \in L^2$, define*

$$Rf = \sum_{n\ge 1} \frac{1}{n} Df_n.$$

Then $Rf \in \operatorname{Dom}\delta$, $\delta Rf = f - \mathbf{E}f$, and

$$\|Rf\|_2^2 = \sum_{n\ge 1} \frac{1}{n}\|f_n\|_2^2 \le \|f\|_2^2.$$

Proof. Orthogonality and the displayed derivative norm show that the field series converges and give its norm formula. For finite chaos sums, Gaussian integration by parts gives $\delta\sum_{n=1}^N n^{-1}Df_n = \sum_{n=1}^N f_n$. The fields and outputs both converge strongly. Since a Hilbert adjoint is closed, the limiting pair belongs to the original adjoint graph. Completeness of Hermite polynomials identifies the scalar limit as $f - \mathbf{E}f$. □

This proof constructs a static Gaussian field. It supplies no time ordering or predictable representative.

Apply R fiberwise in the scenery coordinate, writing the bounded map as R_X. Fubini and the norm bound give a bounded map on the product L^2 space. Approximate a product-space input by finite sums of tensors, use the one-factor adjoint identity, and pass to the original closed partial adjoint graph. Consequently, for every $F \in \mathcal{K}$,

$$R_X F \in \operatorname{Dom} \delta_X, \qquad \delta_X R_X F = F - M, \qquad M = \mathbf{E}_X F, \qquad \|R_X F\|_2 \le \|F\|_2. \tag{7}$$

Here and below a clock-only variable is lifted to the product space when needed. Tensor approximation is legitimate because conditional expectation, the recovery map and the tensor transports are bounded. It does not assert that conditional expectation preserves an arbitrary derivative domain.

Source correspondence: `IBMNativeSourceRecovery`
`fiber_recovery_native_graph; source_recovery_graph; source_recovery_norm`

Proposition 3.2 (Static centered surjectivity)**.** *Every centered $F \in \mathcal{H}$ is an output of $\delta_{\mathcal{H}}$. A bounded choice of preimage is $(R_X F, R_Y M)$, and its squared norm is at most $2\|F\|_2^2$. Conversely, every output of $\delta_{\mathcal{H}}$ is centered.*

Proof. Equation (7) and Proposition 3.1 give ambient outputs $F - M$ and $M - \mathbf{E}F$. Their sum is $F - \mathbf{E}F \in \mathcal{H}$, so Lemma 1.1 gives the asserted intrinsic output. Conditional expectation is a contraction, and hence the sum of the two field norms squared is at most $\|F\|_2^2 + \|M\|_2^2 \le 2\|F\|_2^2$. Finally $1 \in \operatorname{Dom} D_{\mathcal{H}}$ and $D_{\mathcal{H}} 1 = 0$. Testing any adjoint output against 1 gives expectation zero. □

Source correspondence: `IBMCenteredSourceRepresentation`
`centered_reconstruction; range_iff_centered; recovery_pair_bound_sq`

Surjectivity of this static divergence is therefore a different statement from predictable representation in physical IBM time.

4 Actual multitime Fourier observations

Fix a finite index set I, times $r_i \ge 0$, real frequencies a_{ki}, phases β_k, and real weights w_k, where $k \ge 0$. Assume

$$\sum_{k \ge 0} |w_k| < \infty. \tag{8}$$

The uniformly absolutely convergent series

$$F = \sum_{k \ge 0} w_k \cos\left(\beta_k + \sum_{i \in I} a_{ki} Z_{r_i}\right) \tag{9}$$

defines a bounded intrinsic random variable. For $b \in \mathbb{R}^I$ set

$$A(b) = \sum_i |b_i|, \qquad Q_b(x) = \sum_{i,j} b_i b_j \min(|x_i|, |x_j|).$$

The nonnegativity of this expression is essential: signed frequencies cannot be handled by pretending that its individual summands are positive.

Lemma 4.1 (Folded Gram identity)**.** *For every $x \in \mathbb{R}^I$,*

$$Q_b(x) = \left\| \sum_i b_i 1_{(0,|x_i|]} \right\|^2_{L^2(\mathbb{R}_+)} \geq 0.$$

Moreover the actual conditional source mean is

$$M(Y) = \mathbf{E}_X F = \sum_k w_k \cos \beta_k \exp\left(-\tfrac{1}{2} Q_{a_k}((Y_{r_i})_i)\right). \tag{10}$$

Proof. The inner product of $1_{(0,s]}$ and $1_{(0,t]}$ is $\min(s,t)$; expanding the norm proves the first claim. For a fixed clock realization, $\sum_i b_i X_{|Y_{r_i}|}$ is a centered real Gaussian with variance $Q_b((Y_{r_i})_i)$. Its characteristic function at frequency one is $\exp(-Q_b/2)$. Taking the real part after multiplication by $e^{i\beta_k}$ gives the expectation of each cosine. The summable constant $\sum_k |w_k|$ dominates the series, so Fubini or dominated convergence interchanges the source integral with the sum. This also proves $\mathbf{E}_Y M = \mathbf{E}F$. □

Source correspondence: `IBMFoldedSourceCovariance gram; variance_expansion; variance_nonneg`

Source correspondence: `IBMFourierHybridReconstruction source_vacuum; clockMean_expectation`

Each term of the right side of (10) is a continuous function of the finite clock vector. Its modulus is at most $|w_k|$. Uniform convergence therefore makes their sum continuous and bounded. In particular, M lies in the L^2 space generated by the actual Brownian clock, and if every $r_i \leq T$ it is measurable for its raw past at T. Ambient Gaussian measurability alone would not justify either conclusion in a larger Gaussian realization.

5 Strong graph rules at folds and collisions

We use the original closed Gaussian derivative D on the clock factor. Its bounded C^1 chain rule says that bounded ϕ with continuous bounded derivative satisfies $D\phi(f) = \phi'(f)Df$ for $f \in \operatorname{Dom} D$. This is the closed extension of cylindrical differentiation: take a core approximation to f, use bounded slopes for scalar convergence, and dominated multiplier convergence for the derivative. For the latter, split the derivative error as $\phi'(f_n)(Df_n - Df) + (\phi'(f_n) - \phi'(f))Df$; convergence in probability with a uniform bound gives convergence in L^2 against the fixed field. Closedness then gives the original graph identity.

Theorem 5.1 (Absolute values and minima)**.** *If $f, h \in \operatorname{Dom} D$, then $|f|, \min(f,h) \in \operatorname{Dom} D$ and*

$$\begin{aligned} D|f| &= \operatorname{sgn}(f) Df, \\ D\min(f,h) &= \tfrac{1}{2}\{Df + Dh - \operatorname{sgn}(f-h)(Df - Dh)\}. \end{aligned}$$

Here $\operatorname{sgn}(0) = 0$. The respective norm bounds are $\|Df\|_2$ and $\|Df\|_2 + \|Dh\|_2$. No null-level-set hypothesis is required.

Proof. Put $\varepsilon_n = (n+1)^{-1}$, $r_\varepsilon(x) = \sqrt{x^2 + \varepsilon^2}$, and

$$\phi_n(x) = \frac{\sin(\varepsilon_n r_{\varepsilon_n}(x))}{\varepsilon_n}.$$

For each fixed n the function is bounded, continuously differentiable and

$$\phi_n'(x) = \cos(\varepsilon_n r_{\varepsilon_n}(x))\frac{x}{r_{\varepsilon_n}(x)}, \quad |\phi_n'| \le 1, \quad |\phi_n(x)| \le |x| + 1.$$

Pointwise, $\phi_n(x) \to |x|$ and $\phi_n'(x) \to \operatorname{sgn}(x)$, including $x = 0$. The chain rule applies separately for every n. Dominated convergence gives $\phi_n(f) \to |f|$ in L^2, since $f \in L^2$ on a probability space, and $\phi_n'(f)Df \to \operatorname{sgn}(f)Df$ in the field L^2 norm. Closedness proves the absolute-value identity and its norm bound. Now use $\min(f,h) = (f + h - |f - h|)/2$ and linearity of the closed derivative. For its norm, apply the triangle inequality and $|\operatorname{sgn}| \le 1$ to obtain $\frac{1}{2}(\|Df\| + \|Dh\| + \|Df - Dh\|) \le \|Df\| + \|Dh\|$. □

Source correspondence: `GaussianAbsoluteMinimumGraph`
`absolute_graph; minimum_graph`

Lemma 5.2 (Reflected damping)**.** *For $c \ge 0$ and $f \in \operatorname{Dom} D$,*

$$De^{-c|f|} = -ce^{-c|f|}\operatorname{sgn}(f)Df, \qquad \|De^{-c|f|}\|_2 \le c\|Df\|_2.$$

Proof. Use $\psi_n(x) = \exp(-cr_{\varepsilon_n}(x))$. These are bounded by one and have continuous derivatives $-c\psi_n(x)x/r_{\varepsilon_n}(x)$ bounded in modulus by c. The derivatives converge pointwise to $-ce^{-c|x|}\operatorname{sgn}(x)$. Apply the bounded chain rule, scalar and field dominated convergence, and closedness, just as in Theorem 5.1. □

Source correspondence: `GaussianReflectedDampingGraph`
`closed_graph`

These are original first-derivative statements. No second derivative, divergence membership of the resulting field, or physical-time Tanaka formula follows from them.

6 The source mean in the original clock domain

Let $e_r = 1_{(0,r]}$ be the clock Gaussian direction, so $DY_r = e_r$ and $\|e_r\| = \sqrt{r}$. Introduce the actual random direction fields

$$\begin{aligned} v_r &= \operatorname{sgn}(Y_r)e_r, \\ H_{rs} &= \tfrac{1}{2}\{v_r + v_s - \operatorname{sgn}(|Y_r| - |Y_s|)(v_r - v_s)\}, \\ V_b &= \sum_{i,j} b_i b_j H_{r_i r_j}, \qquad C_b = \sum_{i,j} |b_i b_j|(\sqrt{r_i} + \sqrt{r_j}). \end{aligned}$$

The cusp theorem proves the actual graph identities

$$D|Y_r| = v_r, \quad D\min(|Y_r|, |Y_s|) = H_{rs}, \quad DQ_b(Y) = V_b.$$

For $r = s$, the formula gives $H_{rr} = v_r$, including $Y_r = 0$. The pointwise field bounds give

$$\|V_b\|_2 \le C_b \le 2A(b)^2 \sum_i \sqrt{r_i}. \tag{11}$$

For the last inequality, the exact double sum equals $2A(b)\sum_i |b_i|\sqrt{r_i}$, which is at most the stated bound. All sums over I are finite, so these operations are legitimate in the linear closed graph. They cover zero times, signed frequencies, repeated times and an empty observation set.

Set

$$g_b = -\tfrac{1}{2} e^{-|Q_b(Y)|/2} \operatorname{sgn}(Q_b(Y)) V_b.$$

Since $Q_b \geq 0$, Lemma 5.2 gives

$$D e^{-Q_b(Y)/2} = g_b, \qquad \|g_b\|_2 \leq C_b/2. \tag{12}$$

Nonnegativity removes the absolute value in the scalar *value*. The graph output retains the zero-level sign convention; there is no need to silently replace $\operatorname{sgn}(Q_b)$ by one on the set where $Q_b = 0$.

Source correspondence: `IBMMultitimeClockVarianceGraph`
`overlap_graph; variance_graph; varianceCost_bound; damping_graph`

Theorem 6.1 (Completed mean graph)**.** *In addition to* (8)*, suppose*

$$\sum_k |w_k| A(a_k)^2 < \infty. \tag{13}$$

Then the source mean in (10) *belongs to* $\operatorname{Dom} D_Y$ *and*

$$D_Y M = G := \sum_k w_k \cos \beta_k \, g_{a_k}, \qquad \|G\|_2 \leq \left(\sum_k |w_k| A(a_k)^2 \right) \sum_i \sqrt{r_i}.$$

Both the scalar series defining M and this vector series are norm summable.

Proof. The scalar summand is bounded by $|w_k|$. Equations (11) and (12) bound the derivative summand by $|w_k| A(a_k)^2 \sum_i \sqrt{r_i}$. Thus the partial sums converge in both components of the original graph. Closedness proves membership and the output formula; the triangle inequality gives the bound. If representatives are desired, L^1 summability of the pointwise field norms follows from the L^2 estimates on a probability space. Tonelli then gives almost-everywhere absolute convergence to a representative of the same L^2 field. □

Source correspondence: `IBMFourierClockGraph`
`multitime_clock_graph; multitime_clock_domain`

The conclusion concerns M, not F. A smoothing effect of source averaging does not transfer original clock-domain membership back to the observation. The summability condition is sufficient, not claimed necessary or sharp.

7 Predictable representation of the hidden clock mean

Let $\mathcal{F}_t^Y = \sigma(Y_s : 0 \leq s \leq t)$, using the raw Brownian past and its L^2 equivalence classes. Let $\mathcal{P}_Y$ be the closed predictable energy space in $L^2(ds\, d\mathbf{P}_Y)$, obtained by completing adapted step fields. Its original Itô integral

$$J : \mathcal{P}_Y \longrightarrow L^2(\Omega_Y)$$

is an isometry with centered outputs. Let $\mathcal{B}_Y$ be the closed L^2 space generated by the Brownian path, and let P_{pred} be orthogonal projection of direction fields onto $\mathcal{P}_Y$.

Theorem 7.1 (Brownian range and uniqueness)**.** *The range of J is exactly the centered subspace of $\mathcal{B}_Y$. For every $M \in \mathcal{B}_Y$, its unique predictable coefficient is $U = J^* M$, and*

$$JU = M - \mathbf{E}M, \qquad \|U\|_2 = \|M - \mathbf{E}M\|_2 \leq \|M\|_2.$$

Proof. The isometry has closed range. It remains to prove that a dense collection of centered Brownian functionals lies in that range. For deterministic step h of compact support, put

$$E_t(h) = \exp\left(\int_0^t h_s\, dY_s - \frac{1}{2}\int_0^t h_s^2\, ds\right).$$

The elementary Brownian Itô formula gives $E_T(h) - 1 = J(1_{[0,T]}hE_\cdot(h))$. Its field has finite energy because $\mathbf{E}E_s(h)^2 = \exp(\int_0^s h^2)$. These exponentials span a dense subspace of $\mathcal{B}_Y$: if a variable is orthogonal to all of them, condition on any finite vector of independent Brownian increments. In finite Gaussian dimension its Laplace transform after multiplication by the Gaussian density vanishes at every real vector. Cauchy–Schwarz with Gaussian exponential moments makes this transform entire, so successive one-variable uniqueness gives a vanishing Fourier transform and hence the zero conditional variable. An increasing collection of rational-time vectors generates the continuous Brownian path; convergence of conditional expectations then makes the original variable zero. This proves density.

Centering is continuous, so centered exponentials have dense span in the centered Brownian space. Closed range now proves surjectivity onto that space. For any isometry J, $J^*J = 1$ and JJ^* is orthogonal projection onto its range. The orthogonal projection of $M \in \mathcal{B}_Y$ onto the centered subspace is $M - \mathbf{E}M$. This proves the formula. Isometry proves uniqueness and the norm equality. □

Source correspondence: `GaussianBrownianPRP`
`integralRange_eq_centeredBrownianL2; brownian_representation_exists_unique`

The range claim is deliberately restricted to $\mathcal{B}_Y$. It would be false for unrelated Gaussian coordinates in a larger ambient realization. For the source mean (10), membership in $\mathcal{B}_Y$ was established by its actual finite clock-vector formula.

Proposition 7.2 (Support and stability)**.** *If M is measurable for $\mathcal{F}_T^Y$, its coefficient U vanishes after T. For any two Brownian-generated inputs, $\|J^*M - J^*M'\|_2 \le \|M - M'\|_2$.*

Proof. For predictable step fields, independence of future Brownian increments and conditional expectation give $\mathbf{E}(Ju \mid \mathcal{F}_T^Y) = J(1_{[0,T]}u)$. Extend by the integral isometry and contraction of conditional expectation. Apply this identity to U. Since $JU = M - \mathbf{E}M$ is past measurable, $J(1_{(T,\infty)}U) = 0$; the isometry forces the corresponding field to vanish. Finally the adjoint of an isometry has operator norm at most one, proving stability. □

Source correspondence: `IBMFourierClockClarkOcone`
`coefficient_supported; finite_reconstruction; coefficient_stability`

8 Identifying the coefficient by the original derivative

The representation theorem requires only $M \in \mathcal{B}_Y$. The next identity requires membership in the original closed derivative domain.

Theorem 8.1 (Projected derivative)**.** *If $M \in \operatorname{Dom} D_Y \cap \mathcal{B}_Y$, then $J^*M = P_{\mathrm{pred}} D_Y M$. Consequently, under* (8) *and* (13)*, the actual Fourier source mean has predictable coefficient $P_{\mathrm{pred}}G$, where G is the vector series in Theorem 6.1.*

Proof. First take a smooth bounded past cylinder H measurable at time a and the step field $u_s = H1_{(a,b]}(s)$. Its integral is $H(Y_b - Y_a)$. Gaussian integration by parts in the future increment direction gives

$$\mathbf{E}\{MH(Y_b - Y_a)\} = \mathbf{E}\left\{H \int_a^b D_s M \, ds\right\}.$$

The additional product-rule term involving $D_s H$ is zero for $s \in (a, b]$, because every direction generating H is supported in $[0, a]$. This calculation is first for cylindrical M. Approximation in its original derivative graph extends it to all $M \in \operatorname{Dom} D_Y$: the integral side is continuous in scalar L^2, and the right side is continuous in the field L^2 norm, for each fixed bounded step field.

Bounded smooth past cylinders are dense in the past L^2 space, by finite Gaussian conditional approximation. Their predictable step fields are dense in $\mathcal{P}_Y$ by its definition. The integral isometry and Cauchy–Schwarz therefore extend the pairing identity to every $u \in \mathcal{P}_Y$. It reads $\langle J^*M, u\rangle = \langle D_Y M, u\rangle$. Since J^*M lies in $\mathcal{P}_Y$, this is precisely the orthogonal projection characterization. Substitute Theorem 6.1 for the Fourier case. □

Source correspondence: `GaussianBrownianPRP`
`clark_ocone; clark_ocone_representer_unique`

Source correspondence: `IBMFourierClockClarkOcone`
`projected_actual_derivative; stochastic_reconstruction`

For adapted step fields the same calculation proves $\delta_Y u = Ju$ in the original Gaussian adjoint graph. Approximation by steps, isometry and closedness extend this equality to every predictable energy field. This domain assertion will be used in the hybrid theorem.

9 Hybrid reconstruction through the original intrinsic adjoint

For a clock-only predictable field U, write $C(U)$ for its lift to the product space, constant in the scenery coordinate. This is an isometry. Testing on product cylinders and using density gives

$$C(U) \in \operatorname{Dom} \delta_Y, \qquad \delta_Y C(U) = JU. \tag{14}$$

Indeed the clock-only adjoint identity integrates over each scenery coordinate, and finite tensor approximation extends it to the full partial derivative domain. Equivalently, take a clock adjoint graph approximation and transport its constant-scenery tensor through the closed partial graph. No observable or predictable projection is commuted with a derivative.

Theorem 9.1 (Actual Fourier hybrid identity)**.** *Assume only* (8)*, and let* F *be* (9)*,* $M = \mathbf{E}_X F$*, and* $U = J^*M$*. Then*

$$(R_X F, C(U)) \in \operatorname{Dom} \delta_{\mathcal{H}}, \qquad \delta_{\mathcal{H}}(R_X F, C(U)) = F - \mathbf{E}F.$$

For every $f \in \operatorname{Dom} D_{\mathcal{H}}$ *one has the exact duality*

$$\mathbf{E}\{(F - \mathbf{E}F)f\} = \langle R_X F, D_X f\rangle + \langle C(U), D_Y f\rangle.$$

The squared norm of the pair is at most $2\|F\|_2^2$*. If all* $r_i \le T$*, the clock field is supported in* $[0, T]$*. Under* (13) *it is the lift of* $P_{\mathrm{pred}} G$*.*

Proof. The actual source-mean formula makes M Brownian generated. Equation (7), Theorem 7.1 and (14) give the original ambient adjoint outputs

$$\delta_X R_X F = F - M, \qquad \delta_Y C(U) = M - \mathbf{E}_Y M = M - \mathbf{E}F.$$

Their sum is the intrinsic variable $F - \mathbf{E}F$, so Lemma 1.1 removes the final projection on this *sum*. Its proof also gives the duality against every original joint-domain test. The squared field norm is at most $\|F\|_2^2 + \|M - \mathbf{E}M\|_2^2 \le 2\|F\|_2^2$. Proposition 7.2 gives time support. Only the last projected derivative identification uses Theorem 6.1 and therefore the stronger summability hypothesis. □

Source correspondence: `IBMFourierHybridReconstruction`
`partial_graph; clock_mem; clock_value; reconstruction; duality`

The norm estimate here is a direct human deduction from the mapped original recovery bound and Brownian isometry. It is not a new formal declaration or a claim that this joint preimage is unique. A joint divergence may have a kernel even though its clock-only predictable coefficient is unique.

10 A nonzero observable with zero clock mean

Take $T > 0$ and $F = \sin Z_T$, which is a one-term Fourier observation after a phase change. Conditional on Y, the scenery value Z_T is centered Gaussian of variance $|Y_T|$. Symmetry gives

$$M = \mathbf{E}_X \sin Z_T = 0, \qquad U = J^* M = 0.$$

Nevertheless the double-angle identity and the Gaussian characteristic function give

$$\mathbf{E}F = 0, \qquad \mathbf{E}F^2 = \tfrac{1}{2}\left(1 - \mathbf{E}e^{-2|Y_T|}\right) > 0.$$

The strict inequality follows because Y_T is nondegenerate, so $|Y_T| > 0$ almost surely and $e^{-2|Y_T|} < 1$ almost surely. In particular, the entire predictable hidden-clock channel vanishes while F is nonzero. The hybrid theorem instead reads $\delta_{\mathcal{H}}(R_X F, 0) = F$.

Source correspondence: `IBMFourierClockClarkOcone`
`coefficient_zero_of_cos_zero`

Source correspondence: `IBMActualSineDomainObstruction`
`sinePoint_not_constant; sinePoint_not_joint_domain`

The verified source additionally proves that this positive-time sine observation is outside the original joint derivative domain. That separate domain obstruction is not needed in the variance proof above and is not reproved here. It is consistent with our theorem: membership of an *output* in the range of an adjoint does not require it to be in the domain of the operator being adjointed. The paper's claim of complete reconstruction is therefore a claim about an original adjoint output, not point-observation differentiability.

11 Classical context and precise remaining obligations

The Gaussian recovery DN^{-1} on centered chaoses, Brownian predictable representation and the projected-derivative mechanism are classical. Ocone's primary institutional publication record [1] describes the extension of Clark's formula to weakly H-differentiable functionals. This supplies

relevant historical context. Our bounded search inspected that record and abstract, not the complete article; it does not establish priority for any specific IBM formulation or match all analytic hypotheses. The paper claims neither a new Clark–Ocone theorem nor external novelty. Its contribution as a formalization companion is the explicit alignment of actual folded covariance, original cusp graph, completed Fourier mean, predictable clock coefficient and projected intrinsic adjoint.

Three qualifications are mathematical requirements, rather than choices of terminology. First, $\mathcal{F}_t^Y$ is the hidden clock's Brownian past. The observed past $\sigma(Z_s : s \le t)$ is a different sigma algebra. Nothing above identifies U as predictable for that observed past. Second, $R_X F$ is a static Gaussian source field; it is not constructed as a physical-time predictable IBM integrand. Third, even the full observable projection P used in the adjoint identity is not a causal projection. Dropping any of these distinctions would change the theorem.

To obtain an observed-filtration physical-time reconstruction, one would still need a specified observed predictable integral space, an original closed integral realizing the desired physical prescription, admissibility of the proposed coefficients in that space, and a proof that its centered residual vanishes in the stated topology. The sine test shows why success for conditional source means cannot discharge those obligations. Existing Simpson identities and native-grid weak-limit obstructions retain their own domains and topologies; no transfer to arbitrary partitions, nonsmooth physical-time formulas or a predictable representation property is asserted.

References

[1] D. Ocone, *Malliavin's calculus and stochastic integral representations of functionals of diffusion processes*, Stochastics **12** (1984), 161–185. doi:10.1080/17442508408833299. Primary institutional record and abstract: Rutgers publication record.

[2] Project Stardust, *OperatorDerivativeRewrite*, frozen verified source commit identified in Section 1. The accompanying `theorem-map.json` gives declaration-level permanent source links.

[3] D. Nualart, *Malliavin Calculus and Normal Approximations*, primary author notes, Propositions 5.2 and 5.4, equation (30), printed pages 39–41. `https://nualart.ku.edu/sites/nualart/files/documents/updated/Course_Malliavin_Calculus_accessible.pdf`.